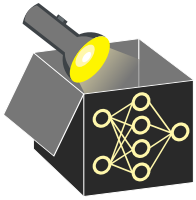
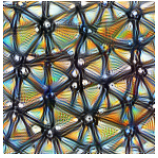
# Interpretable Machine Learning



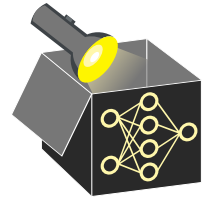
## Post-hoc Methods for Neural Networks



**Learning goals**

- Interpretability in neural networks
- Landscape of interpretability
- The difference between feature visualization and feature attributions

# NEURAL NETWORKS AS COMPLEX ML MODELS



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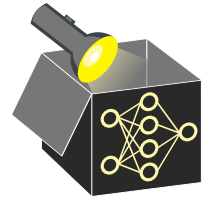

- Neural networks are over parameterised
  - Vision models and Language models routinely have > millions of params
  - Sometimes #parameters > #input instances
  - Which and how do the features, parameters, training instances contribute towards the final decision ?

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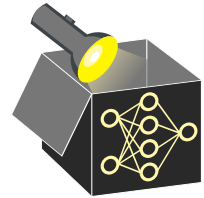

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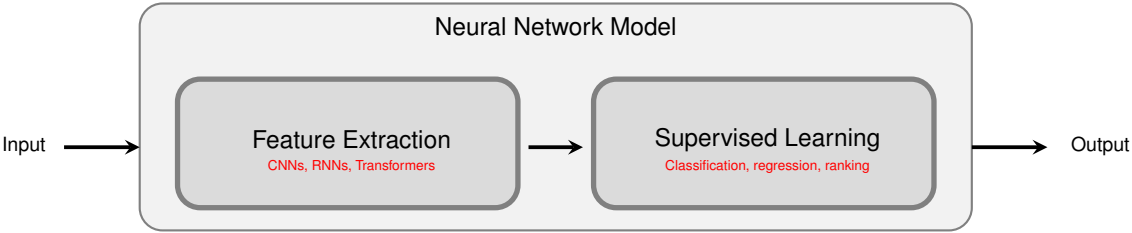

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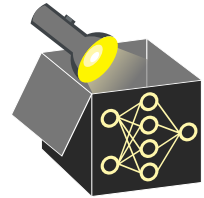
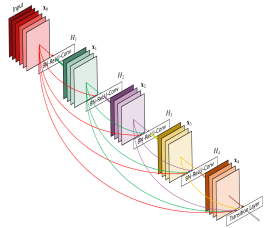
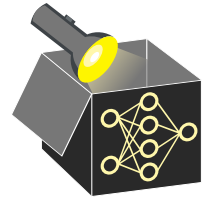


- Neural networks are compositional and non-linear systems
  - The success of neural networks is due to their depth
    - Depth results in compositional behaviour
  - Non-linearity between layers helps capture non-linear relationships

- Depth and non-linearity leads to lack of interpretability

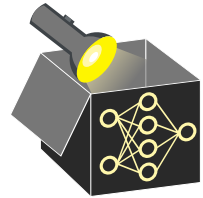


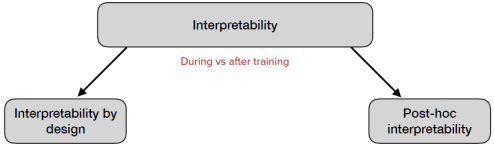
# MODEL-SPECIFIC INTERPRETABILITY



- What types of neural models are out there ?
  - For vision: Convolutional Neural Nets
  - For language, speech: Recurrent Neural Nets, Transformer Models
  - For recommendation systems, ranking: Factorization-based Models, Embeddings models
- Each of the domains have their challenges and have developed specific approaches for interpretability
  - We will focus on first principles that can be applied to most models
  - We will discuss adaptations to each data modality as and when required

# INTERPRETABILITY LANDSCAPE IN NEURAL NETWORKS





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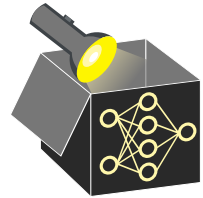

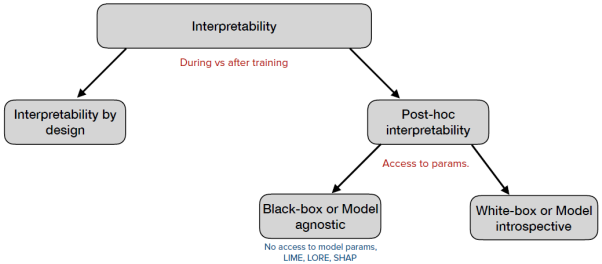

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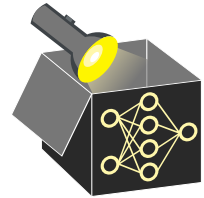

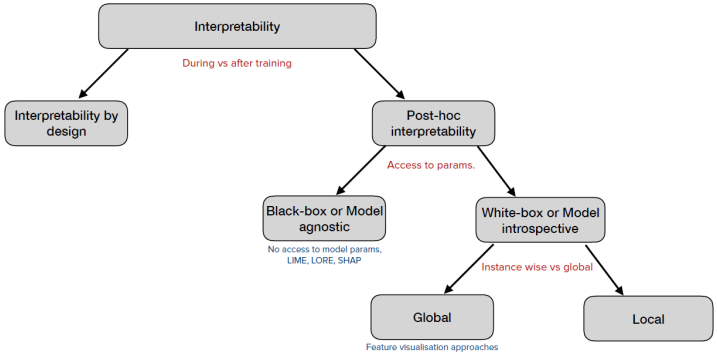

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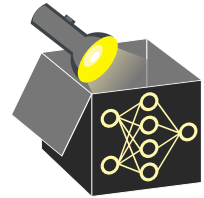

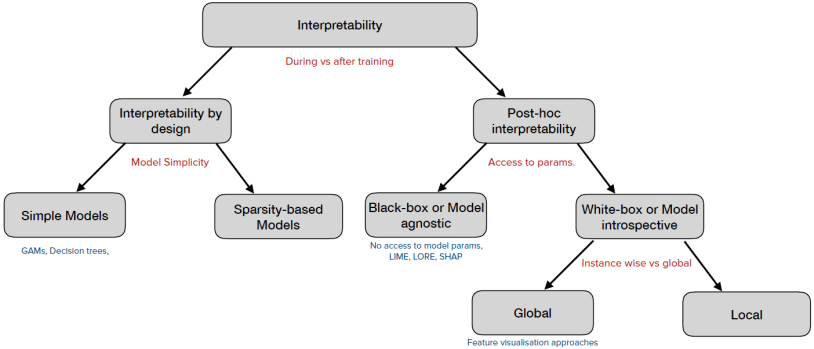

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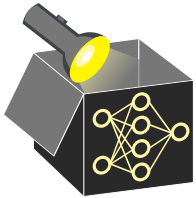
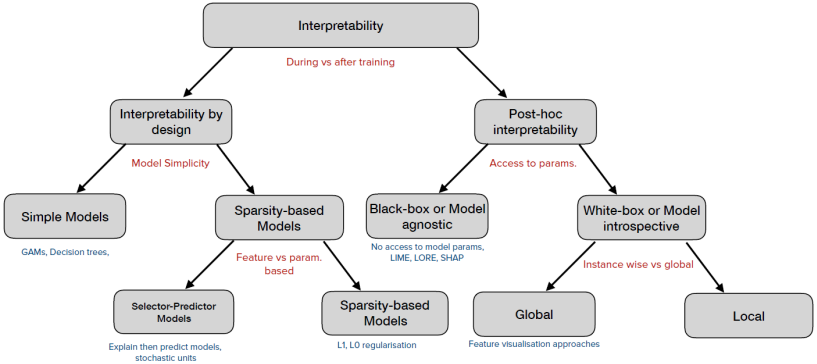
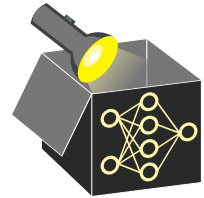




# HOW CAN WE INTERPRET NEURAL MODELS ?



- Feature visualization: Visualizing components of the neural networks
  - Activations of neurons
  - Attention values
  - Gradient flow
- Feature attributions: relevant input features
  - Which input features are responsible for the given decision ?
  - Sensitivity analysis using gradient-based methods
  - Using black-box methods like LIME, SHAP, etc.

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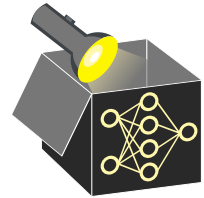

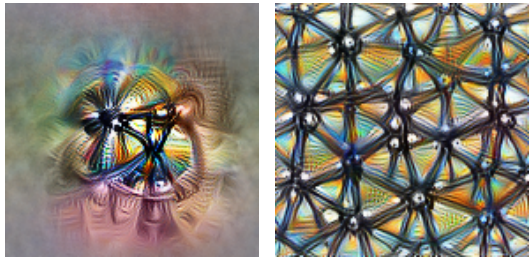

**Figure:** Feature visualization: Visualizing components of the neural networks

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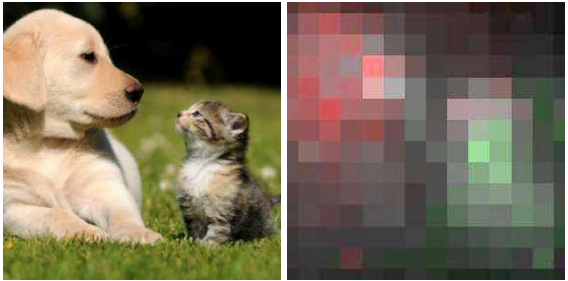

**Figure:** Feature attributions: relevant input features